

4. Let $g_{\mu\nu}$ be a rank $(0,2)$ tensor. In a basis, one can regard the components $g_{\mu\nu}$ as elements of an $n \times n$ matrix, so that one may define the determinant $g = \det(g_{\mu\nu})$. How does g transform under a change of basis?

5*. Use the Leibniz rule to derive the formula for the Lie derivative of a 1-form ω , valid in any coordinate basis:

$$(\mathcal{L}_X \omega)_\mu = X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu$$

[Hint: consider $(\mathcal{L}_X \omega)(Y)$ for a vector field Y .] Show that the Lie derivative of a $(0,2)$ tensor g is

$$(\mathcal{L}_X g)_{\mu\nu} = X^\rho \partial_\rho g_{\mu\nu} + g_{\mu\rho} \partial_\nu X^\rho + g_{\rho\nu} \partial_\mu X^\rho$$

For a p -form η , define $\iota_X \eta$ to be the $(p-1)$ -form that results from contracting a vector field X with the first index of η . Show that for a 1-form ω ,

$$\mathcal{L}_X \omega = i_X(d\omega) + d(i_X \omega)$$

6. Let ω be a p -form and η a q -form. Show that the exterior derivative satisfies the properties

- $d(d\omega) = 0$
- $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge d\eta$
- $d(\varphi^* \omega) = \varphi^*(d\omega)$ where $\varphi : M \rightarrow N$ for some manifolds M and N

7. The exterior derivative of $\omega \in \Lambda^p(M)$ can be defined as

$$\begin{aligned} d\omega(X_1, \dots, X_{p+1}) &= \sum_{j=1}^{p+1} (-1)^{j-1} X_j(\omega(X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_{p+1})) \\ &\quad + \sum_{j < k} (-1)^{j+k} \omega([X_j, X_k], X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_{k-1}, X_{k+1}, \dots, X_{p+1}) \end{aligned}$$

In a coordinate basis $\omega = \frac{1}{p!} \omega_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$, use this definition to determine the components of $d\omega$ in the cases $p = 1$ and $p = 2$.

8. A three-sphere can be parameterized by Euler angles (θ, ϕ, ψ) where $0 < \theta < \pi$, $0 < \phi < 2\pi$, $0 < \psi < 4\pi$. Define the following 1-forms

$$\sigma_1 = -\sin \psi d\theta + \cos \psi \sin \theta d\phi, \quad \sigma_2 = \cos \psi d\theta + \sin \psi \sin \theta d\phi, \quad \sigma_3 = d\psi + \cos \theta d\phi$$

Show that $d\sigma_1 = \sigma_2 \wedge \sigma_3$ with analogous results for $d\sigma_2$ and $d\sigma_3$.